PQC DAILY STUDY SERIES

FALCON: A Deep Dive into the Post-Quantum Digital Signature Algorithm

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Introduction to FALCON

FALCON (Fast-Fourier Lattice-based Compact Signatures over NTRU) is a post-quantum digital signature algorithm selected by the U.S. National Institute of Standards and Technology (NIST) for standardization. It is designed to be a secure and efficient replacement for classical digital signature algorithms like RSA and ECDSA, which are vulnerable to attacks from large-scale quantum computers.

FALCON is a lattice-based cryptosystem, which means its security is based on the difficulty of solving certain mathematical problems on lattices. Specifically, it is built on the NTRU (N-th degree Truncated polynomial Ring Units) lattice, which allows for compact and efficient signatures.

How FALCON Works

FALCON's signature generation process involves the following steps:

1. **Key Generation:** A key pair (public and private) is generated. The private key is a pair of polynomials with small coefficients, and the public key is derived from the private key.
2. **Hashing:** The message to be signed is hashed using a cryptographic hash function (e.g., SHA-3).
3. **Signature Generation:** The signature is generated using the private key and the hash of the message. This process involves a technique called "hash-and-sign" and uses a fast and efficient algorithm based on the Fast Fourier Transform (FFT).
4. **Signature Verification:** The signature is verified using the public key, the message, and the signature itself.

FALCON's Applicability to Internet Protocols

FALCON's efficiency and small signature size make it an excellent candidate for use in various internet protocols, including:

- **TLS (Transport Layer Security):** FALCON can be used for digital signatures in TLS certificates, ensuring the authenticity of web servers in a post-quantum world.
- **SSH (Secure Shell):** FALCON can be used to authenticate servers and clients in SSH connections, protecting against quantum attacks.
- **VPNs (Virtual Private Networks):** FALCON can be used to secure the key exchange and authentication in VPN protocols like IPsec and WireGuard.
- **Code Signing:** FALCON can be used to sign software and firmware, ensuring their integrity and authenticity.

FALCON vs. Other PQC Digital Signature Algorithms

Algorithm	Key Size (bytes)	Signature Size (bytes)	Security Level
FALCON-512	1,793	666	1
ML-DSA-44	1,312	2,420	2
SLH-DSA-SHA2-128s	32	7,856	1

Daily Quiz

1. What does the acronym FALCON stand for?
2. What is the underlying mathematical problem that provides security for FALCON?
3. Name two internet protocols where FALCON can be used.